

Baron Li

baronjk2027@gmail.com (858) 692-6003

EDUCATION

The Bishop's School, La Jolla, CA

August 2023 - Now

Relevant Coursework:

Multi-variable Calculus, Advanced Honors Physics, Calculus BC, Honors Biology, Honors Chemistry, Honors Computer Science Algorithms

SAT: 1560

AP exams: Computer A (5), Calculus BC (5), Calculus AB (5), Chemistry (5), US History (5).

To take in 2026: Physics C Mechanics, Physics C E&M, Biology

PROFESSIONAL EXPERIENCE

Research Intern, UC San Diego,

June 2025 - Now

Researched using Liquid Xenon as a liquid scintillation crystal in PET scanners:

- Studied the design and mechanics of conventional PET scanners with Dr. Ni.
- Designed and simulated PET systems on point sources using Opengate 10, an open source library used for particle simulations.
- Reconstructed medical images using the CasTor Project reconstruction and compared the results of LXe and LYSO scintillation crystals.
- Published in the Journal of Instrumentation, April 2026.
- Participated in the 72nd Greater San Diego Science & Engineering Fair and CSEF.

Research Intern, UC Davis,

June 2025 - Now

Researched the design of a 170 GHz notch filter for fusion reactor diagnostics:

- Studied fusion plasma diagnostics with Professor Zhu from UC Davis
- Designed a notch filter to protect the receiver sensors from the 170 GHz waves emitted by cyclotron heating.
- Used CST Studio Suite 2025 to simulate and optimize filter performance
- Fabricated the actual filter and used it on a real fusion reactor.
- Registered for the 26th High Temperature Plasma Diagnostic Conference hosted by the MIT Plasma Science & Fusion Center from June 7-11, 2026.
- 1st-author paper submitted to RSI in June 2026.

Polygence Research Program

August 2024 - January 2025

Researched the possible effects of a hypothetical Planet 9 on Pluto:

- Based on a theoretical model of universal gravity, an analysis of the orbit and eccentric change of Dwarf Planet Pluto due to hypothetical Planet 9.
- Provided a feasible approach to identify the existence of Planet 9, potentially if given ground-truth data.
- Published in Curieux Academic Journal, April 2025.
- Attended Polygence Symposium of Rising Scholars, March 2025.

App Development

June 2024 - December 2024

Developed an online App, "Smart Trails."

- The app was developed based on NOLS Outdoor Expeditions' experience.
- Developed to help hikers register their hiking trips before entering trails, by setting a time limit on the App.
- The App will automatically call emergency systems (first contact the user's trusted contact, then contact authorities if further action is required) if nobody stops the timer.
- Potentially help people when they get lost or have emergencies on hiking trails.

EXTRACURRICULARS

Founder and President, SkyGazers Club in The Bishop's School

August 2025 - Now

- Founded a new club in TBS for monitoring incoming astronomy events.
- Study astrophysics and try to predict when skywatcher events will occur
- A mix of theory and practice: problem solving, presentation of discoveries, and debating, etc.
- Study the universe through different lenses to develop a lifelong sense of wonder about the universe.
- Completed a detailed document for the club mission and event calendar. Initiated.

Founder and President, Beamline for Schools (BL4S) in The Bishop's School

October 2025 - Now

- Founded a team of 6 to participate in an event organized by CERN, the European Laboratory for Particle Physics
- Discussed every week, and came up with a final experiment proposal to utilize the facility at CERN.
- Submitted to BL4S for review.

President, Maker's Club in The Bishop's School

August 2025 - Now

- Made things bi-weekly during lunch break.
- Hosted engineering competitions between high-school students, i.e., bridges, buildings etc..

NOLS Outdoor Expedition

July 2024 - August 2024

- Attended a 3-week trip to Wyoming
- Learned various wilderness survival skills - setting up camp, identifying good camping spots, emergency first aid, etc.
- Leadership training- lead a group of 7 people hiking in the wilderness for a few days to survive both on and off-trail

Independent Study

April 2025 - Now

- Read book: Physics and Philosophy: the Revolution in Modern Science
- Completed Coursera course: Philosophy and the Sciences: Introduction to the Philosophy of Physical Sciences
- Attended conference of Foundations of Physics, UC Irvine, June 16-18, 2026
- Created personal website

Research Foundations Program, Non-trivial

March 2026-May 2026

- Chosen for and completed a highly selective five-week research program focused on identifying and analyzing important global challenges.
- Developed core research and critical thinking skills, explored high-impact problem areas, and produced original research directions through guided weekly work, workshops and events, and independent inquiry.

Sailing Varsity team at TBS

July 2024 - Now

- Practice at Mission Bay Yacht Club twice a week
- Attended the national Rose Bowl Regatta at Long Beach in the silver fleet. Finished in 9th place.

Science Olympiad Core team member at TBS

August 2024 - Now

- Meet-ups with event partners to exchange event materials and conduct studies.
- In 2025 SO, won 1st place in "Helicopter" and 4th in "Astronomy."

Robotics club at TBS

August 2024 - Now

- Weekly practice for the soccer project.
- Design and tune the autonomous navigation algorithms.
- Participated in the national competition RoboCup Junior Soccer in May 2026, won 2nd place.

Competitive Coding Club

May 2023 - Feb 2025

- Works on coding problems with CodeForces/USACO/LeetCode
- Weekly meetings and discussions on algorithm development, such as DFS/BFS, Dijkstra, dynamic programming, etc.
- Attended USACO competitions since 2023: qualified silver in February 2024 and gold in December 2024.

MITx Online Learning Course

August 2023 - Now

- Introduction to Mechanics: Kinematics and Dynamics

- Calculus 1A: Differentiation
- Calculus 1B: Integration

COMMUNITY SERVICES

President, SD Girls in STEAM Non-profit Organization & hope4kids Club

June 2025 - Now

- Worked with Casa de Amistad to help kids in need of education
- Hosted UCSD lab tours for Casa de Amistad girls to spread job and educational awareness
- Maintained the website for the organization
- Created 4 branches for local middle and high schools
- Co-hosted a fundraising event with hope4kids organization. Collected over \$10000 for Casa de Amistad.

Branch President, Youth Care Club

July 2019 - Now

- Organize and lead the YCC 92130-care project monthly to collect near-expiring food from stores and distribute to low-income families in the neighborhood.
- Went back to a small town in China to teach local unprivileged kids English for 2 weeks
- Online tutoring for successive summers for students in middle school, high school, and college in China
- Volunteered at other activities: C2K Computer clean up, food banks, and many crane-folding events for Joey's Wings Foundation to inspire children with cancer, etc.
- Fundraised for House of China's December Nights, and collected over \$9000

PUBLICATIONS

- Li, B (first author), Ma, Y, Ni, K., "A New Concept of Liquid Xenon Time Projection Chamber for Medical Imaging", Journal of Instrumentation (JINST), Vol. 21, Apr. 2026, C04019, DOI: 10.1088/1748-0221/21/04/C04019 (LIDINE 2025 Proceedings)
Also available at arXiv:2603.25974
- Baron Li (first author) et al., "Design of a 170 GHz Waveguide-Based ENOCH Notch Filter for Burning Plasma Diagnostics Protection." Submitted to Review of Scientific Instruments (RSI) in June 2026.
- Rahman, K., Li, X., Sun, P., Himes, L., Domier, C., Lai, H., Xu, X., Li, B., Zhu, Y., "Design of the Image Rejection Filter for Fusion Millimeter-wave Diagnostics." Submitted to Journal of Fusion Engineering and Design (FED) in Feb. 2026.
- Baron Li (first author), Sophia Kressy, "Influence of Hypothetical Planet 9 on Dwarf Planet Pluto", Curieux Academic Journal, pgs. 409-415, Part 1, Issue 49, April 2025.

AWARDS

72nd Greater San Diego Science & Engineering Fair

First Award winner in Physics and Astronomy. CSEF qualifier.

3-time American Invitational Mathematics Exam (AIME) Qualifier

- Qualified with Distinction (top 5%)

The United States of America Computing Olympiad (USACO)

- Gold-level in Dec 2024

The President's Volunteer Service Award

- Bronze in 2024
- Gold in 2020

Level 10 Certificate of Merit, Piano (with State Honors)

Non-trivial Research Foundations Program Scholar